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EDUCATION

University of California San Diego

Bachelor of Science in Electrical Engineering

Bachelor of Science in Mathematics-Computer Science

La Jolla, CA

Sep 2024 - June 2028

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EXPERIENCE

Triton Droids - Vice President of Engineering

La Jolla, CA | Jan 2025 – Present

- Architecting the software and systems stack for our second generation humanoid competing in RoboCup. Defined interfaces across perception, locomotion, high level decision making, communications, and low level control.
- Developing the lower body locomotion and control stack for our first generation humanoid, 1.3m, 31kg, 20 DOF, implementing gait generation and joint level control toward stable bipedal walking on physical hardware.
- Derived per joint PD gains, friction coefficients, and armature inertias from RobStride actuator specifications and validated actuator dynamics in Isaac Lab to improve simulation fidelity for locomotion and sim to real deployment.
- Lead software, electrical, and mechanical engineering teams within a 100-member robotics club, coordinating development milestones, cross subsystem integration, and technical documentation across two humanoid generations.
- Designed and integrated system level requirements spanning Jetson/ESP32 compute, CAN motor control, IMU sensing, power distribution, battery management, safety layers, and serviceability.

PROJECTS

OpenRECAP - Self Improving Vision Language Action Policy

Python, Pytorch, VLAs, MuJoCo, LeRobot, Reinforcement Learning

- Implemented an offline self improving pipeline for SmolVLA, using autonomous rollouts, value estimation, and advantage labeled trajectory fine tuning; replaced the original large VLM value model with a lightweight state-based MLP to run the full pipeline on a single GPU.
- Built an end to end MuJoCo environment for the SO-101 integrating teleoperated behavior cloning in simulation, along with autonomous rollouts, reward labeling, value estimation, and iterative policy fine tuning.
- Compared sparse/dense rewards and KL regularization across eight training iterations. Dense rewards improved success from 50% to 60% and increased success for the most difficult position from 20% to 51%.

Humanoid Robot Fighting - Motion Imitation Learning

Python, MuJoCo, MJLab, PPO, RSL-RL

- Built a motion processing pipeline for the EngineAI T800 humanoid, mapping NPY/BVH boxing and martial arts motion onto a floating base and 25 actuated joints, with 50 Hz interactive replay and 1080p rollout rendering.
- Converted 30 Hz mocap into 50 Hz full-body reference trajectories using forward kinematics, quaternion SLERP, kinematic tree mapping, and finite difference velocity estimation.
- Adapted MJLab's PPO tracking stack to the T800, tracking 14 body segments with position, orientation, and velocity rewards plus joint limit, action rate, and self collision penalties. Trained policies capable of reproducing punch/kick motions, qualifying in the top 32 for the first round of the Ultimate Robot Knockout League.

Vision Language Action Models for Manipulation

Python, LeRobot, Action Chunking Transformers

- Fine-tuned and compared $\pi 0.5$ (VLA), ACT, and SmolVLA on cube stacking. $\pi 0.5$ was the only policy to complete the full task, achieving 10% success versus 0% for ACT and SmolVLA under varied cube positions.
- Fine-tuned $\pi 0.5$ for 50K training steps and diagnosed real world deployment failures caused by camera pose, lighting, and teleoperator distribution shift, improving repeatability with rigid 3D printed camera mounts and controlled lighting.
- Trained across H100 GPUs using VESSL AI, Google Colab, and SDSC. Tracked experiments with Weights & Biases and published checkpoints to Hugging Face.

SKILLS

Languages & Tools: Python, C++, MATLAB, PyTorch, LeRobot, MuJoCo, Isaac Lab, HuggingFace, Git, Linux/Ubuntu, Onshape, EagleCAD

Models & Methods: $\pi 0.5$, ACT, SmolVLA, PPO, SAC, Flow Matching, Behavior Cloning, Teleoperation, CAD/Mechanical Design